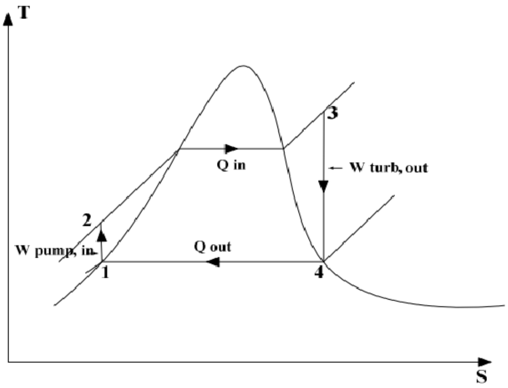
ID: Solution .

Habib University

EE-437 Renewable energy systems – Fall 2025

Time = 40 minutes **Final Exam Part 2** **Max Points: 20**

Question 1 [3 points]: Draw the rankine cycle on the T-S diagram and explain how it differentiates from the carnot cycle?



Type of cycle:
Carnot – Theoretical; Rankine – Practical.

Heat addition:
Carnot – Isothermal; Rankine – Constant pressure.

Heat rejection:
Carnot – Isothermal; Rankine – Constant pressure.

Evaporation:
Carnot – Fully isothermal; Rankine – Has separate evaporation stage.

Compression:
Carnot – Wet vapor compression; Rankine – Liquid water compression (easy).

Expansion:
Both isentropic, but Rankine may start with superheated steam.

Temperature variation:
Carnot – Constant; Rankine – Varies during heat addition/rejection.

Efficiency:
Carnot – Maximum theoretical; Rankine – Lower, but realistic.

Feasibility:
Carnot – Impractical; Rankine – Widely used in power plants.

T–S diagram:
Carnot – Rectangle; Rankine – Curved loop.

Question 2 [2 points]: Explain zenith and azimuth.

Altitude: The angle between the sun's ray and horizon.

Zenith: 90 - Altitude

Azimuth: The compass direction of the Sun along the horizon, measured clockwise from North (0°) through East (90°), South (180°), and West (270°).

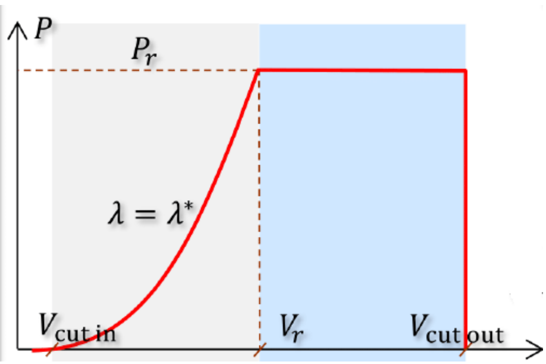
Question 3 [2 points]: Explain how solar radiation changes with latitude and longitude at a specific time

Latitude: Radiation angle changes with increasing latitude, reducing radiation as latitude increases. Furthermore, difference between North and South latitudes are of seasons.

Longitude: Different longitudes will have different times of the day, so one longitude can be at noon and getting full radiation while one could be in the dark.

Question 4 [3 points]: Sketch the graph for power vs wind speed for a typical wind turbine. Make sure to mark V_{cutin} , V_r and V_{cutoff}

What are the four distinct regions in the power vs speed graph?



Region 1: V< Vcut in : No energy production

Region 2: Vcut in < V < Vr : Constant TSR strategy

Region 3: Vr < V < V cutout: Constant power

Region 4: V> V Cutout: No energy production

Question 5 [2 points]: Why does hydropower have a much larger share of electric power worldwide compared to solar and wind (4 reasons)?

Hydro is a more mature technology and has been here for over a century.

Hydro is more reliable and predictable.

Hydropower is cheaper than wind and solar

Hydro provides more benefits than just electricity.

Question 6 [1 point]: What factors contribute to the efficiency of a pumped storage plant? (2 factors)

Height difference between the two reservoirs

Efficiency of turbine/generator

Efficiency of pump/motor

Question 7 [1 point]: Why do biomass thermal plants have lower efficiency than coal power plants? (2 reasons).

Low temperature of combustion

Higher water content in fuel

Question 8 [1 point]: Flywheels are used to even out frequencies on the grid level. Why are flywheels more suitable for this operation compared to something like lithium-ion batteries?

Grid stabilization requires high power for very short pulses. Li-Ion would have severe ageing issues. (Any 1 reason is enough)

Question 9 [1 point]: What elements exist in an LFP and MNC cell? Example (Elements in LCO cell are lithium, cobalt and oxygen)

LFP: Lithium, Iron, Phosphorous, Oxygen (any 2)

MNC: Lithium, Manganese, Nickel, Cobalt (any 3)

Question 10 [1 point]: List any 3 reasons a BMS is needed for lithium-ion battery packs.

Sensing

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Protection

Interface

Performance management

SOC and SOH estimation

Cell balancing (Any 3)

Question 11 [1 points]: Differentiate/ define DSO, TSO and IPP.

DSO: Distribution of electric supply in cities

TSO: Transmission of electric supply intercity and even between countries

IPP: Independent power producer of electric supply

Question 12 [1 point]: How can changing a solar panel's orientation help the grid?

By moving to southwest orientation more energy is produced during sunset when energy is in higher demand. Different orientations give smoother daily production.

Question 13 [1 point]: How does a low wind turbine help the grid?

Low speed turbines give more output when winds are slow compared to a classical turbine and less power when winds are high. This gives a smoother output of power.